Classifying Shooting Incident Fatality Using Machine Learning

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Abstract

Shooting incidents represent a significant public safety concern in urban environments. Understanding the factors that influence whether a shooting incident results in a fatality is critical for developing effective prevention strategies and improving emergency response planning. This project focuses on analyzing historical shooting incident data to identify patterns associated with fatal and non-fatal outcomes.

Using the NYPD Shooting Incident dataset, a comprehensive data science workflow was implemented, including data preprocessing, exploratory data analysis, feature engineering, statistical validation, and machine learning model development. Various temporal, spatial, and demographic features were analyzed to understand their relationship with incident fatality.

Multiple machine learning models were implemented and compared, including Logistic Regression, Support Vector Machine, Decision Tree, Random Forest, Gradient Boosting, and XGBoost. Model performance was evaluated using cross-validation and several metrics such as Precision, Recall, F1 Score, and ROC-AUC to ensure robust evaluation in the presence of class imbalance.

Among the evaluated models, XGBoost demonstrated the most balanced performance and was selected as the final predictive model. The results indicate that shooting incident fatality is influenced by a combination of temporal factors, geographic location, and demographic attributes.

The insights derived from this analysis can assist law enforcement agencies and public safety organizations in developing data-driven strategies for targeted policing, resource allocation, and preventive interventions aimed at reducing fatal shooting incidents.

Step 1 :Executive Summary

Urban public safety remains a critical challenge in many metropolitan areas. Understanding the factors that contribute to fatal outcomes in shooting incidents can assist law enforcement agencies in developing targeted intervention strategies. This project focuses on analyzing historical shooting incident data to identify patterns associated with fatal and non-fatal outcomes.

Using the NYPD Shooting Incident dataset, a comprehensive data analysis and machine learning pipeline was developed to classify incidents as fatal or non-fatal. The project involved extensive data preprocessing, exploratory data analysis, feature engineering, and statistical validation to identify meaningful patterns in the data.

Multiple machine learning algorithms were evaluated, including Logistic Regression, Support Vector Machine, Decision Tree, Random Forest, Gradient Boosting, and XGBoost. After comparative evaluation using cross-validation and multiple performance metrics such as Precision, Recall, F1-Score, and ROC-AUC, the XGBoost classifier demonstrated the best overall performance.

The final model achieved moderate predictive capability and provided insights into the temporal, spatial, and demographic factors influencing fatal shooting incidents. These insights can support data-driven decision-making for law enforcement agencies by highlighting high-risk conditions and enabling more effective allocation of resources.

This project demonstrates how machine learning techniques can be applied to real-world public safety data to generate actionable insights while maintaining awareness of ethical considerations and data limitations.

Step 2 : Problem Statement

Shooting incidents represent a major public safety concern in urban environments. While many shooting incidents result in non-fatal injuries, some lead to fatalities, which have severe social and economic consequences for communities.

Understanding the conditions that increase the likelihood of fatal outcomes can help authorities develop preventive strategies and improve emergency response planning. However, identifying such patterns using traditional analysis methods can be challenging due to the complexity of the factors involved.

The primary objective of this project is to develop a machine learning model capable of classifying shooting incidents as fatal or non-fatal based on historical incident data. By analyzing various attributes such as incident time, location, victim demographics, and perpetrator characteristics, the model aims to identify factors associated with higher fatality risk.

The results of this analysis can assist law enforcement agencies and public safety organizations in prioritizing resources, designing targeted interventions, and improving overall situational awareness.

Step 3: Project Objectives

The main objectives of this project include:

- Analyze historical shooting incident data to understand patterns related to fatal and non-fatal outcomes.
- Perform comprehensive data preprocessing and exploratory data analysis to ensure data quality and reliability.
- Identify key factors that influence fatality outcomes in shooting incidents.
- Develop and compare multiple machine learning models for incident classification.
- Evaluate model performance using appropriate metrics suitable for imbalanced classification problems.
- Select the most suitable model based on predictive performance and interpretability.
- Provide actionable insights that can support public safety decision-making.

Step 4: Dataset Description

The dataset used in this project is the **NYPD Shooting Incident Data (Historic)** dataset, which contains records of shooting incidents reported in New York City over multiple years.

The dataset includes detailed information about each incident, such as:

- Incident date and time
- Borough and precinct location
- Victim demographics
- Perpetrator demographics
- Geographic coordinates of the incident
- Indicator of whether the incident resulted in a fatality

These attributes allow for the analysis of temporal, spatial, and demographic patterns associated with shooting incidents.

The target variable for this project is **STATISTICAL_MURDER_FLAG**, which indicates whether a shooting incident resulted in a fatality.

Target variable categories:

- True – Fatal incident

- False – Non-fatal incident

This dataset provides a valuable foundation for applying machine learning techniques to understand the conditions under which shooting incidents lead to fatalities.

Step 5: Data Understanding

Understanding the dataset is a critical first step in any data science project. Before applying preprocessing techniques or building machine learning models, it is important to examine the dataset structure, data types, and overall data quality.

The NYPD Shooting Incident dataset contains historical records of shooting incidents reported across New York City. Each record represents an individual incident and includes information about the incident location, victim demographics, perpetrator demographics, and the fatality outcome.

Initial exploration of the dataset was performed using functions such as `head()`, `tail()`, `info()`, and shape to gain a basic understanding of the dataset.

The dataset consists of multiple attributes representing different aspects of each incident, including:

- Temporal attributes such as incident date and time
- Spatial attributes such as borough, precinct, and geographic coordinates
- Victim demographic attributes
- Perpetrator demographic attributes
- Incident classification indicating whether the shooting resulted in a fatality

The target variable for this project is **STATISTICAL_MURDER_FLAG**, which indicates whether the incident resulted in a fatal outcome.

Understanding the dataset structure helped identify potential data quality issues such as missing values, inconsistent data types, and redundant attributes. These observations guided the subsequent preprocessing steps.

Step6 : Data Preprocessing

Data preprocessing is an essential stage in the machine learning pipeline, as raw datasets often contain missing values, redundant information, and inconsistent formats that can negatively impact model performance.

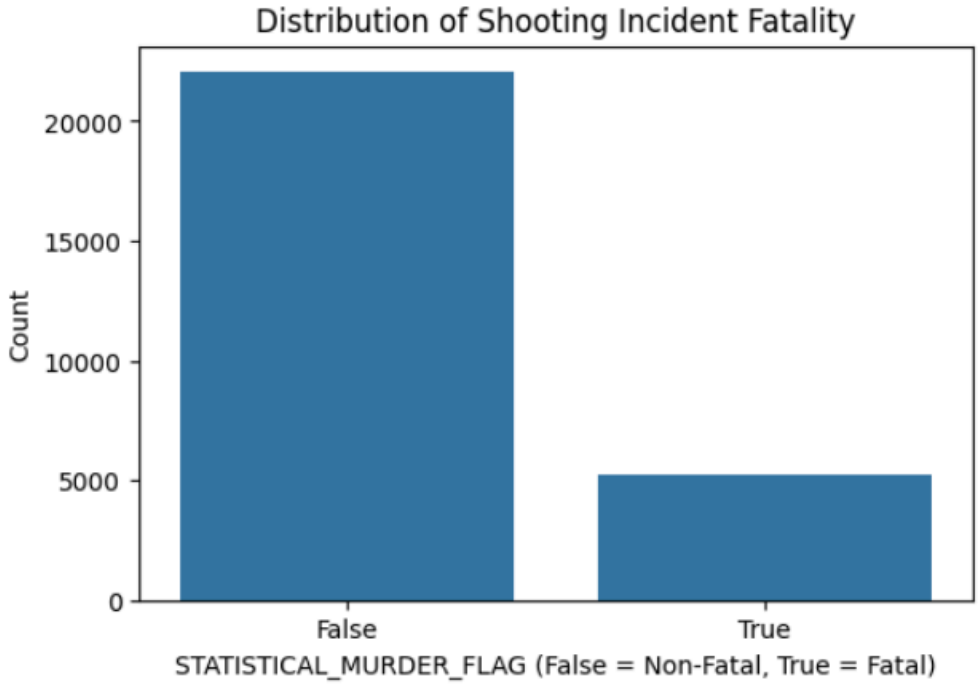
The preprocessing process involved several steps to ensure that the dataset was clean, structured, and suitable for machine learning model development.

Target Variable Analysis

The first step in preprocessing involved analyzing the distribution of the target variable, **STATISTICAL_MURDER_FLAG**, which represents whether a shooting incident resulted in a fatality.

The analysis revealed that the dataset is **class imbalanced**, with non-fatal incidents significantly outnumbering fatal incidents. Such imbalance can lead to biased models if not properly addressed during model training and evaluation.

Understanding this imbalance was important for selecting appropriate evaluation metrics and for implementing strategies such as class weighting and resampling during the modeling phase.



Missing Value Analysis

A comprehensive missing value analysis was performed to determine the extent of incomplete data across the dataset.

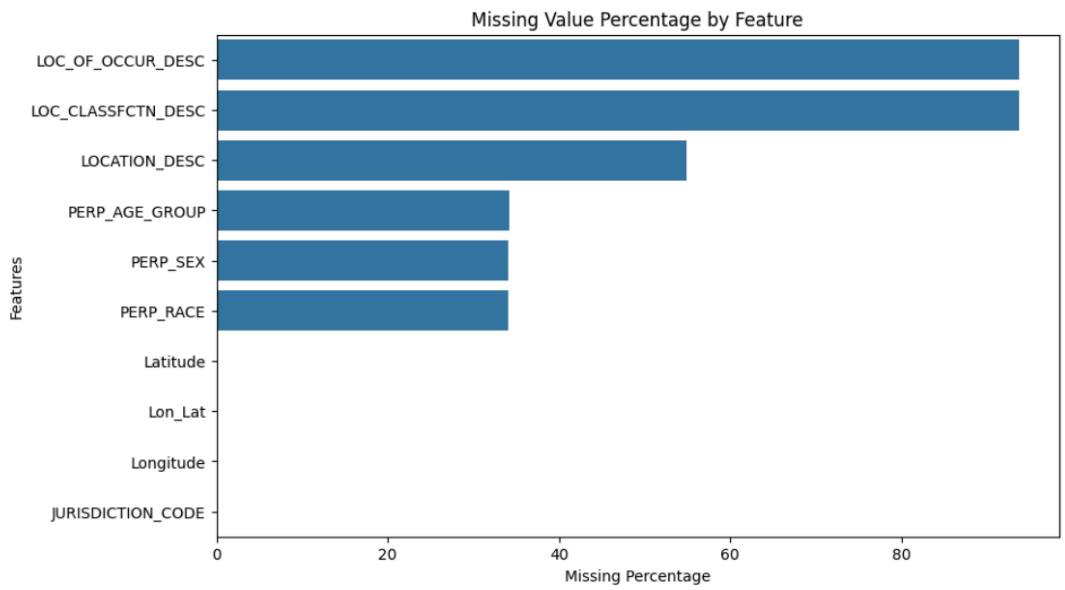
The analysis revealed that certain features contained a very high percentage of missing values. For example:

- LOC_OF_OCCUR_DESC
- LOC_CLASSFCTN_DESC
- LOCATION_DESC

These columns contained a substantial number of missing entries, which reduced their reliability for predictive modeling.

Features with extremely high missing value percentages were removed from the dataset, as imputing such large proportions of missing data could introduce noise and bias into the model.

Other features with moderate missing values were carefully evaluated and handled using appropriate preprocessing strategies.



Unique Value Analysis

Unique value analysis was performed to understand the cardinality of each feature in the dataset.

This analysis helped identify:

- High-cardinality features such as geographic coordinates
- Identifier-like attributes
- Features with limited categorical levels

Understanding feature cardinality was important for selecting appropriate encoding techniques later in the modeling pipeline.

For example, features with high cardinality were handled using **frequency encoding**, while low-cardinality categorical features were encoded using **one-hot encoding.**

Duplicate Data Check

Duplicate records can introduce bias into machine learning models and distort statistical analysis.

A duplicate check was performed to ensure data integrity.

The analysis showed that the dataset did not contain duplicate rows, confirming that each record represents a unique shooting incident.

Data Cleaning

Based on the results of the missing value and unique value analyses, several data cleaning actions were performed.

Columns with extremely high missing value percentages were removed to prevent unreliable information from affecting the model.

Additionally, redundant or less informative attributes were eliminated to reduce noise in the dataset.

These steps helped simplify the dataset while retaining the most relevant features for analysis and modeling.

Data Type Conversion

During initial exploration, several features were identified with incorrect or inconsistent data types.

To ensure proper data handling and enable feature extraction, data types were converted as necessary.

For example:

- Incident date columns were converted to **datetime** format to allow temporal feature extraction.
- Incident time columns were converted to **time-based** formats for hourly analysis.
- Certain categorical features were converted to appropriate categorical data types to improve memory efficiency and analysis clarity.

Correct data typing ensured that the dataset could be properly processed in subsequent analysis stages.

Step 7 : Feature Engineering

Feature engineering plays a crucial role in improving the predictive capability of machine learning models. Raw features often do not fully capture the underlying patterns present in the data. Therefore, additional features were created to represent temporal patterns and contextual relationships that may influence fatality outcomes in shooting incidents.

Several new features were derived from existing attributes to enhance the dataset.

Temporal Feature Extraction

Temporal information from the incident date and time was used to generate additional features that capture patterns related to when incidents occur.

The following features were extracted:

- **YEAR –** Represents the year in which the shooting incident occurred.
- **MONTH –** Indicates the month of the incident.
- **MONTH_NAME –** Human-readable month label used for visualization and analysis.
- **DAY_OF_WEEK –** Numeric representation of the day of the week.
- **DAY_NAME –** Day label used for easier interpretation during exploratory analysis.
- **HOUR –** Hour of the day when the incident occurred.

These features help identify temporal patterns such as seasonal trends, weekday versus weekend incidents, and time-of-day risk factors.

Context-Based Feature Creation

In addition to temporal features, several contextual indicators were created to capture conditions that may influence fatal outcomes.

The following engineered features were introduced:

- **IS_NIGHT –** Indicates whether the incident occurred during nighttime hours.
- **IS_WEEKEND –** Identifies incidents that occurred on weekends.
- **IS_SUMMER –** Indicates whether the incident occurred during summer months.
- **IS_ELDERLY_VICTIM –** Identifies whether the victim belongs to an elderly age group.
- **IS_ELDERLY_PERP –** Indicates whether the perpetrator belongs to an elderly age group.
- **NIGHT_WEEKEND –** Interaction feature capturing incidents that occurred during nighttime on weekends.

These engineered features allow the model to capture contextual patterns that may not be immediately visible in the raw dataset.

Step 8 : Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand the underlying patterns within the dataset and to identify relationships between features and fatal shooting outcomes.

EDA helps uncover trends, anomalies, and correlations that can inform feature selection and model development.

The analysis was divided into several categories.

Target Variable Analysis

The distribution of the target variable, **STATISTICAL_MURDER_FLAG**, was analyzed to understand the proportion of fatal and non-fatal shooting incidents.

The analysis showed that non-fatal incidents represent the majority of cases, while fatal incidents occur less frequently. This indicates the presence of class imbalance, which is an important consideration during model training and evaluation.

Understanding this imbalance helped guide the choice of evaluation metrics and model training strategies.

Temporal Pattern Analysis

Temporal features were analyzed to understand when shooting incidents are more likely to occur.

The analysis revealed patterns related to:

- Yearly distribution of incidents
- Monthly variations in shooting activity
- Day-of-week patterns
- Hourly trends in incident occurrence

These patterns provided valuable insights into how shooting incidents vary across different time periods.

Categorical Feature Analysis

Categorical features such as victim demographics and perpetrator characteristics were analyzed to understand their distribution and potential relationship with fatal outcomes.

The analysis focused on features such as:

- Victim age group
- Victim race
- Perpetrator age group
- Perpetrator race
- Incident location attributes

This analysis helped identify demographic patterns associated with shooting incidents and potential risk factors influencing fatality.

Spatial Analysis

Spatial attributes such as borough, precinct, and geographic coordinates were analyzed to understand the geographic distribution of shooting incidents.

Mapping and spatial visualization techniques were used to identify areas with higher incident concentration.

Understanding spatial patterns helps identify regions that may require targeted policing or increased resource allocation.

Bivariate Analysis

Bivariate analysis was conducted to examine the relationship between individual features and the target variable.

This analysis helped identify which features are more strongly associated with fatal shooting outcomes.

Comparisons were made between fatal and non-fatal incidents across multiple features, including:

- Temporal attributes
- Victim demographics
- Perpetrator demographics
- Location attributes

This step provided insights into the factors that may contribute to fatal outcomes.

Step 9 : Statistical Validation

To validate the relationships observed during exploratory analysis, statistical testing techniques were applied.

Correlation Analysis

Correlation analysis was performed on numerical features to identify potential relationships between variables.

Highly correlated variables were carefully evaluated to avoid redundancy and reduce multicollinearity in the dataset.

Chi-Square Test

For categorical features, the Chi-Square statistical test was used to determine whether there is a significant relationship between each feature and the target variable.

Features with statistically significant relationships were considered more informative for predictive modeling.

This statistical validation step helped ensure that the selected features contribute meaningful information to the machine learning models.

Step 10 : Feature Encoding

Before training machine learning models, categorical features in the dataset were transformed into numerical representations, as most machine learning algorithms require numerical input.

Two encoding strategies were used based on the cardinality of the features.

High Cardinality Encoding

Some features such as PRECINCT and LOCATION_DESC contained a large number of unique categories. Applying traditional one-hot encoding to these variables would significantly increase the dimensionality of the dataset.

To address this issue, frequency encoding was applied to these features. Frequency encoding replaces each category with its relative frequency within the dataset. This approach preserves useful information while preventing excessive feature expansion.

Low Cardinality Encoding

Categorical features with a limited number of unique values were encoded using one-hot encoding.

The following features were encoded using this method:

- PERP_AGE_GROUP
- PERP_SEX
- PERP_RACE
- VIC_AGE_GROUP
- VIC_RACE
- JURISDICTION_CODE

One-hot encoding creates binary indicator variables for each category, allowing machine learning models to effectively interpret categorical data.

Step 11 : Feature Selection

After feature engineering and encoding, the dataset contained several derived and encoded features. A feature selection process was performed to retain only the most relevant attributes for model training.

The final feature set included:

- Spatial features such as latitude and longitude
- Temporal features including year, month, day of week, and hour
- Engineered contextual indicators
- Encoded demographic attributes
- Frequency encoded location features

This step ensured that the models were trained using informative and meaningful predictors.

Step 12 : Model Development

Multiple machine learning algorithms were implemented and compared to determine the most effective approach for predicting shooting incident fatality.

The following models were evaluated:

- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree
- Random Forest
- Gradient Boosting
- XGBoost

Using multiple algorithms allowed a comprehensive comparison between linear models, tree-based models, and ensemble methods.

Train-Test Split

The dataset was divided into training and testing subsets to evaluate model performance on unseen data.

A train-test split strategy was used where:

- 80% of the data was used for model training
- 20% of the data was reserved for testing

Stratified sampling was applied to ensure that the class distribution of fatal and non-fatal incidents remained consistent across both training and testing datasets.

Cross Validation

To ensure robust model evaluation and reduce the risk of overfitting, **Stratified K-Fold Cross Validation** was applied during the model training process.

This technique divides the training data into multiple folds and trains the model repeatedly using different subsets of the data. The final performance score is calculated as the average across all folds.

Cross-validation provides a more reliable estimate of model performance compared to a single train-test split.

Step 13 : Handling Class Imbalance

The target variable exhibited a significant class imbalance, with non-fatal incidents occurring much more frequently than fatal incidents.

Class imbalance can lead to biased models that favor the majority class. To address this issue, two approaches were evaluated.

Class Weight Adjustment

In the first approach, class weights were applied to penalize misclassification of the minority class (fatal incidents). This method allows the model to give more importance to fatal cases during training.

SMOTE Resampling

The second approach involved applying **SMOTE (Synthetic Minority Over-sampling Technique)** to generate synthetic samples of the minority class in the training dataset.

Model performance was compared before and after applying SMOTE to evaluate whether resampling improved predictive performance.

Step 14 : Model Comparison

All models were evaluated using multiple performance metrics to ensure a balanced assessment of classification performance.

The evaluation metrics used included:

- Accuracy
- Precision
- Recall

- F1 Score
- ROC-AUC Score

Since the dataset is imbalanced, metrics such as `Precision`, `Recall`, and `F1 Score` were considered more informative than accuracy alone.

A comparative analysis of all models was performed to determine the best-performing algorithm.

Among the evaluated models, the **XGBoost classifier** demonstrated the most balanced performance across key metrics, particularly in terms of F1 Score and ROC-AUC.

Step15 : Hyperparameter Tuning

To further optimize the performance of the selected model, hyperparameter tuning was conducted using **RandomizedSearchCV.**

RandomizedSearchCV explores different combinations of hyperparameters across a defined search space, allowing the identification of improved model configurations.

The tuned model was evaluated and compared against the baseline configuration to determine whether tuning provided significant performance improvements.

Step16 : Final Model Selection

Based on the model comparison and evaluation process, XGBoost was selected as the final model for this project.

The selection was based on several factors:

- Strong performance across multiple evaluation metrics
- Robust handling of complex feature interactions
- Ability to capture non-linear relationships within the data

XGBoost demonstrated the best balance between predictive performance and generalization capability.

Step 18 : Model Evaluation

The final selected model, XGBoost, was evaluated using multiple performance metrics to assess its effectiveness in predicting whether a shooting incident results in a fatal outcome. Because the dataset exhibits class imbalance between fatal and non-fatal incidents, evaluation focused not only on accuracy but also on metrics such as precision, recall, F1 score, and ROC-AUC.

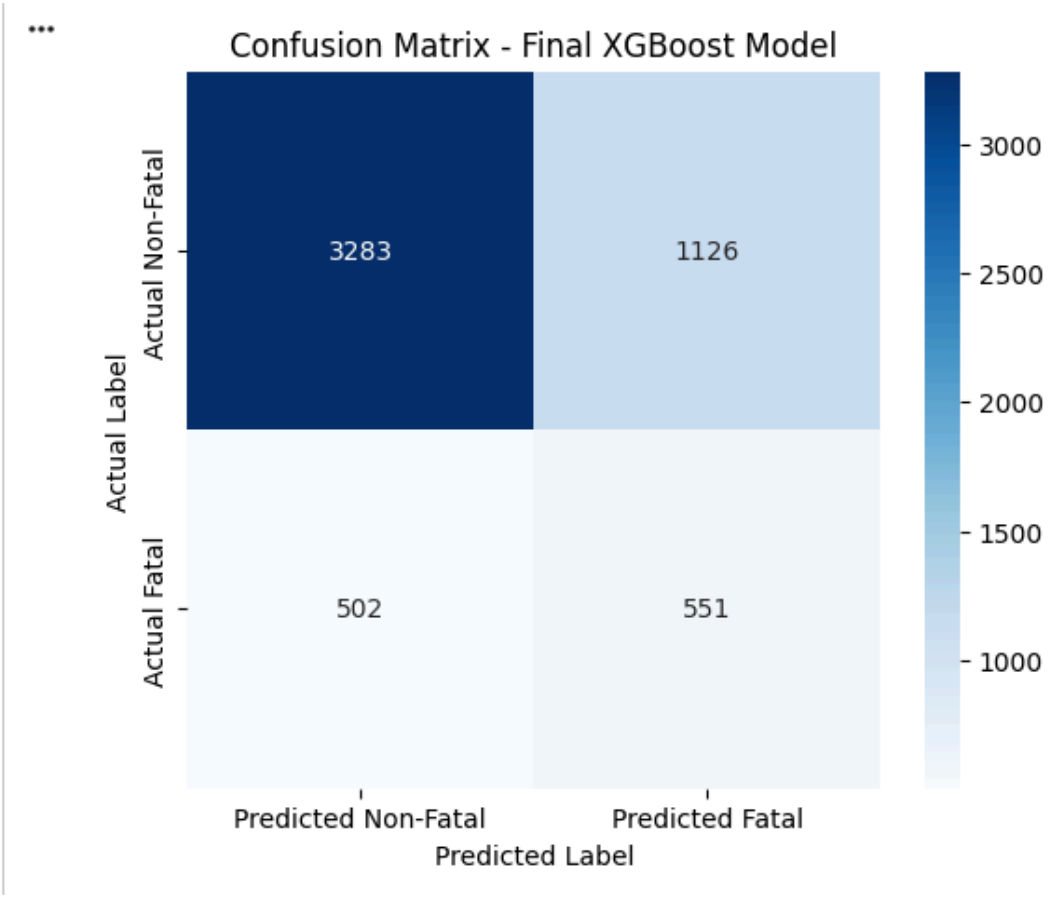
The accuracy of **approximately 66%** indicates that the model correctly classifies a majority of incidents overall. However, accuracy alone is not sufficient for evaluating performance in imbalanced classification problems.

The **precision score of 0.308** suggests that among the incidents predicted as fatal by the model, around 30% were actually fatal. This relatively moderate precision indicates the presence of false positives, where non-fatal incidents are sometimes predicted as fatal.

More importantly, the **recall score of 0.606** shows that the model successfully identifies around **60% of actual fatal incidents**. From a public safety perspective, recall is a critical metric because failing to identify fatal incidents (false negatives) could have serious implications.

The **F1 score of 0.409** reflects the balance between precision and recall. Given the complexity of the dataset and the inherent unpredictability of real-world violent incidents, this level of performance is considered reasonable for a predictive model built on historical incident data.

The **ROC-AUC score of 0.703** indicates that the model has a good ability to distinguish between fatal and non-fatal incidents across different classification thresholds. A value above 0.70 suggests that the model performs significantly better than random guessing.

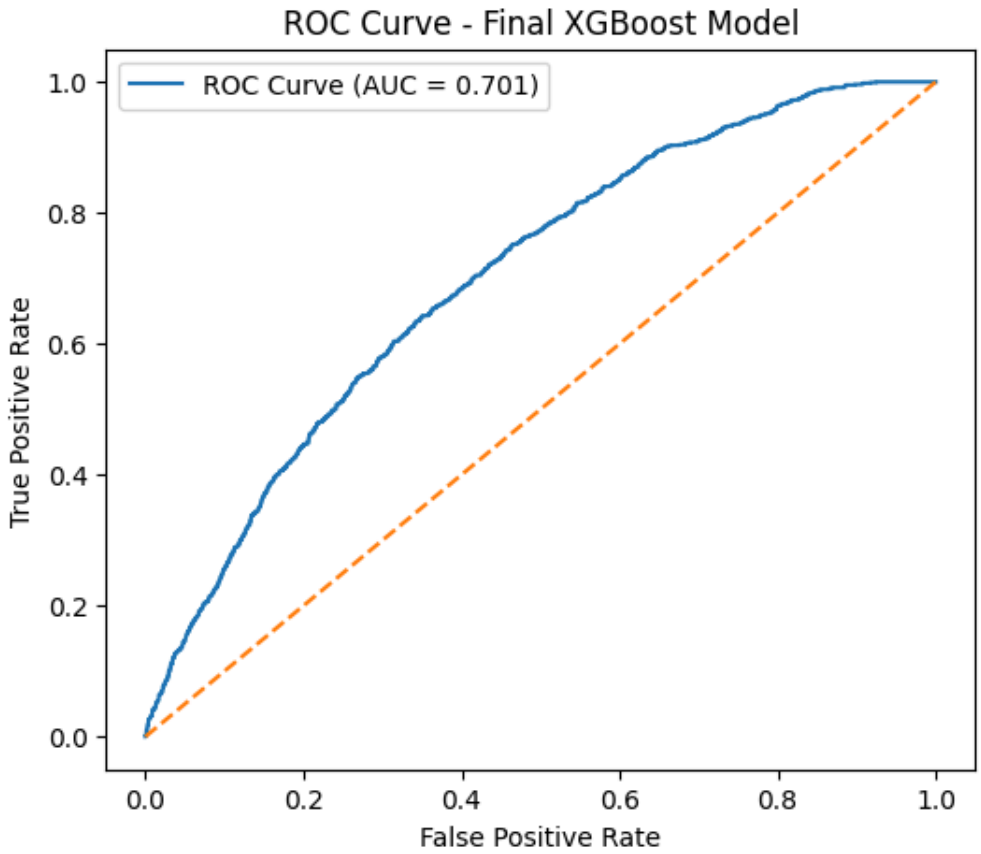


From this matrix we observe:

- **3283** incidents were correctly classified as non-fatal (True Negatives).
- **551** incidents were correctly identified as fatal (True Positives).
- **1126** incidents were incorrectly predicted as fatal even though they were non-fatal (False Positives).
- **502 fatal incidents were missed by the model** and predicted as non-fatal (False Negatives).

Although some misclassification occurs, the model demonstrates a reasonable ability to detect fatal incidents while maintaining acceptable overall performance.

ROC Curve Interpretation



The ROC curve illustrates the trade-off between the **True Positive Rate (Recall)** and **False Positive Rate** across different classification thresholds.

The curve remains well above the diagonal reference line, confirming that the model performs significantly better than random classification. The **ROC-AUC value of 0.701** indicates that the model has a good ability to differentiate between fatal and non-fatal incidents.

This suggests that the selected model captures meaningful patterns within the data and can be used as a decision-support tool for analyzing shooting incident risk factors.

Summary of Model Performance

Overall, the XGBoost model demonstrates a **balanced trade-off between detecting fatal incidents and maintaining overall classification accuracy.** While the model is not perfect, its performance is consistent with expectations for real-world public safety data, where incident outcomes are influenced by many unpredictable factors.

The model's ability to correctly identify a majority of fatal incidents, combined with its acceptable ROC-AUC performance, supports its use as the final predictive model in this project.

Step 19 : Model Interpretability

To understand the factors influencing the prediction of fatal shooting incidents, feature importance analysis was performed using the XGBoost model. Feature importance helps identify which variables contribute most significantly to the model's decision-making process.

The feature importance plot highlights the top predictors used by the model to classify whether a shooting incident resulted in a fatality or not.

The analysis indicates that **perpetrator demographic attributes** play a significant role in predicting fatal outcomes. In particular, categories related to **perpetrator age group**, especially cases where the perpetrator age is unknown, appear as the most influential predictors in the model. This suggests that incomplete demographic information may still carry meaningful signals related to incident severity.

Perpetrator sex variables also appear among the top important features, indicating that demographic attributes associated with the perpetrator contribute to the classification of fatal incidents.

Victim demographic characteristics also contribute to the model. For example, features related to victim age group, particularly victims under the age of 18, appear among the influential predictors. This suggests that certain victim age groups may be associated with different fatality risk patterns.

Temporal variables also influence the prediction process. **The year of the incident** appears as one of the important features, indicating that fatality patterns may vary over time.

Location-based features also contribute to the model's predictions. The **location frequency feature**, derived from the location description, helps capture the relative frequency of incidents occurring in specific locations. This suggests that certain locations may have different fatality risk levels.

Additionally, contextual features such as the **night and weekend interaction feature (NIGHT_WEEKEND)** appear among the top predictors. This indicates that incidents occurring during late hours on weekends may carry different risk characteristics compared to other time periods.

Overall, the feature importance analysis demonstrates that fatal shooting incidents are influenced by a combination of **demographic characteristics, temporal factors, and spatial context**. These insights highlight the complex nature of violent incidents and emphasize the importance of analyzing multiple dimensions when studying public safety data.

Understanding these influential features can help support data-driven decision making for law enforcement agencies by identifying conditions and contexts where incidents may have

a higher likelihood of resulting in fatalities.

Top 15 Feature Importances from the XGBoost Model

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9        IS_ELDERLY_VICTIM    0.015969
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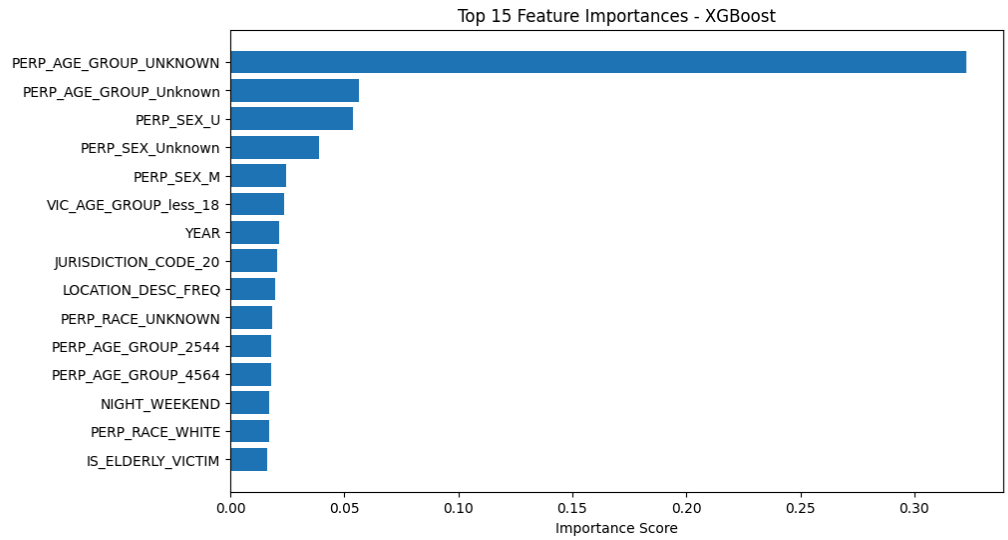

Step 19 : Business Interpretation & Insights

The machine learning model and exploratory analysis provided several insights into patterns associated with shooting incidents and fatal outcomes.

Temporal Patterns

Analysis of temporal features revealed that shooting incidents tend to follow certain time-based patterns. Incidents occurring during nighttime hours and weekends showed relatively higher risk levels compared to daytime incidents. These patterns suggest that certain time periods may require increased monitoring and preventive policing strategies.

Geographic Patterns

Spatial analysis indicated that shooting incidents are concentrated in specific precincts and boroughs. These high-incident areas represent potential hotspots that may benefit from targeted policing efforts and enhanced surveillance.

Demographic Factors

Victim and perpetrator demographic characteristics also showed meaningful patterns. Certain age groups appeared more frequently in incident records, indicating demographic segments that may require focused community outreach or intervention programs.

Contextual Risk Indicators

Engineered features such as nighttime indicators and weekend interaction effects helped highlight contextual conditions associated with shooting incidents. These conditions can inform law enforcement agencies about situations that may carry increased risk.

Overall, the analysis demonstrated that fatal shooting incidents are influenced by a combination of temporal, spatial, and demographic factors, rather than a single isolated variable.

Step 20 Recommendations for Public Safety

Based on the insights derived from the data analysis and machine learning model, several recommendations can be proposed to support public safety initiatives.

Targeted Resource Allocation

Law enforcement agencies may consider allocating additional patrol units and emergency response resources during time periods identified as higher risk, such as nighttime hours and weekends.

Geographic Hotspot Monitoring

Areas with higher concentrations of shooting incidents can be prioritized for proactive policing strategies, surveillance systems, and community safety programs.

Community Engagement Programs

Demographic patterns observed in the data suggest that community-focused intervention programs may help address underlying social factors associated with violent incidents.

Data-Driven Decision Making

The predictive model developed in this project can support data-driven policing strategies by identifying conditions under which fatal outcomes are more likely to occur. Such insights can assist decision-makers in prioritizing preventive measures.

Step 21 : Project Limitations

While this project provides useful insights into shooting incident patterns, several limitations should be acknowledged.

Data Quality Limitations

Certain features contained missing values or incomplete records, which limited the availability of some potentially useful information.

Limited Feature Scope

The dataset primarily contains incident-level information and lacks broader contextual variables such as socioeconomic indicators, environmental conditions, or law enforcement response times.

Class Imbalance

The dataset exhibits class imbalance, with non-fatal incidents significantly outnumbering fatal incidents. Although techniques such as class weighting and resampling were evaluated, class imbalance remains a challenge for predictive modeling.

Geographic Scope

The dataset represents incidents from a specific city. As a result, findings may not generalize to other regions with different demographic or socioeconomic conditions.

Step22 : Future Improvements

Several opportunities exist to further enhance this project in future work.

Incorporating Additional Data Sources

Integrating additional datasets such as socioeconomic indicators, weather data, or neighborhood-level statistics could improve model performance and provide deeper insights.

Advanced Modeling Techniques

Future work could explore advanced machine learning and deep learning techniques to capture more complex patterns in the data.

Real-Time Prediction Systems

The model developed in this project could potentially be integrated into a real-time monitoring system that assists public safety agencies in identifying high-risk situations.

Improved Model Interpretability

Techniques such as SHAP (SHapley Additive Explanations) can be further applied to provide deeper insights into how specific features influence model predictions.

Step23 : Conclusion

This project demonstrated how machine learning techniques can be applied to historical shooting incident data to analyze patterns associated with fatal outcomes.

Through a structured workflow involving data preprocessing, exploratory analysis, feature engineering, statistical validation, and model comparison, a predictive model was developed to classify shooting incidents as fatal or non-fatal.

Among the evaluated models, XGBoost showed the best overall performance and was selected as the final model.

The analysis revealed that fatal shooting incidents are influenced by a combination of temporal, spatial, and demographic factors. These findings highlight the potential value of data-driven approaches in supporting public safety decision-making.

Although the predictive performance is moderate due to the complexity of real-world incidents and dataset limitations, the project successfully demonstrates the application of machine learning to generate actionable insights from public safety data.

The results of this study can contribute to improved understanding of incident patterns and support strategies aimed at reducing fatal outcomes in shooting incidents.

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